MAGNETIC DETECTION & HALL LOGIC

Curiosity Kickoff

- What happens when a magnet comes near?
- Can it detect invisible force?
- What happens when magnet goes away?

Concept Connection

A Magnetic Detection & Hall Logic system uses a Hall effect sensor to detect magnetic fields and process the signal using logic circuits. When a magnet is present, the sensor generates a signal that controls outputs like LEDs or alarms, enabling smart and reliable magnetic sensing applications.

1. HallEffect (Magnet) Sensor :

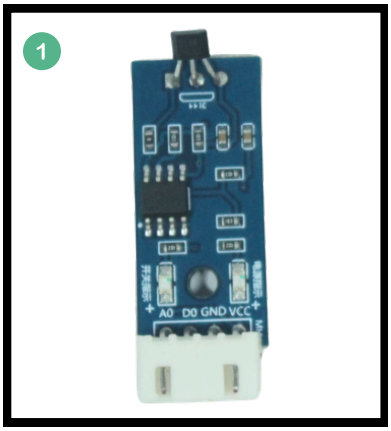
Detects the presence of a magnetic field,Outputs a voltage when a magnet is near.
Pinout:3 pins - VCC: Power, GND: Ground & OUT: Digital output

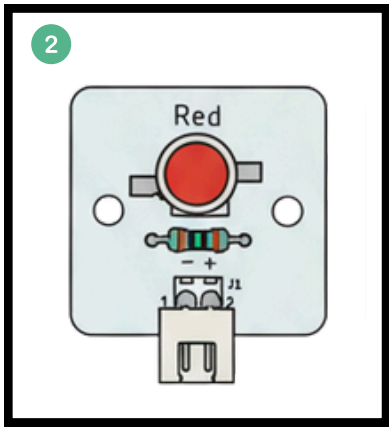
2. Watt SMD LED

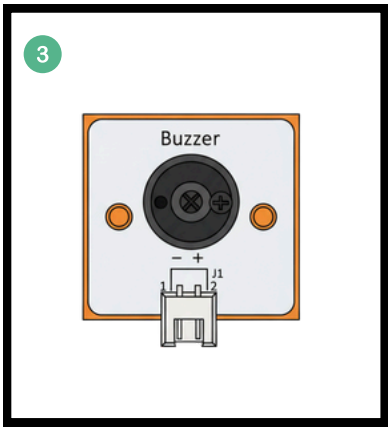
I'm flat and powerful. Just one of me can light up your whole table like a torch.I don't like wasting energy, so I shine super bright without getting too hot,You've already met me — I live in your mobile flashlight!

3.Buzzer

A sound-making device that works when electricity makes a thin metal plate vibrate rapidly inside. The vibration creates buzzing or beeping sounds. Some buzzers can even change frequency to play simple tunes or melodies.







Beginner: Magnetic Detect

Class Explanation

A Magnetic Detect system uses a magnetic sensor (like a reed switch or Hall sensor) to detect the presence of a magnetic field. When a magnet comes near, the sensor triggers an output such as an LED or alarm, useful in security systems and door sensors.

Materials Required

- Arduino with attached shield
- 1 × HallEffect (Magnet) Sensor
- 1 × white 1 Watt SMD LED on breakout board
- RMC wires
- USB cable

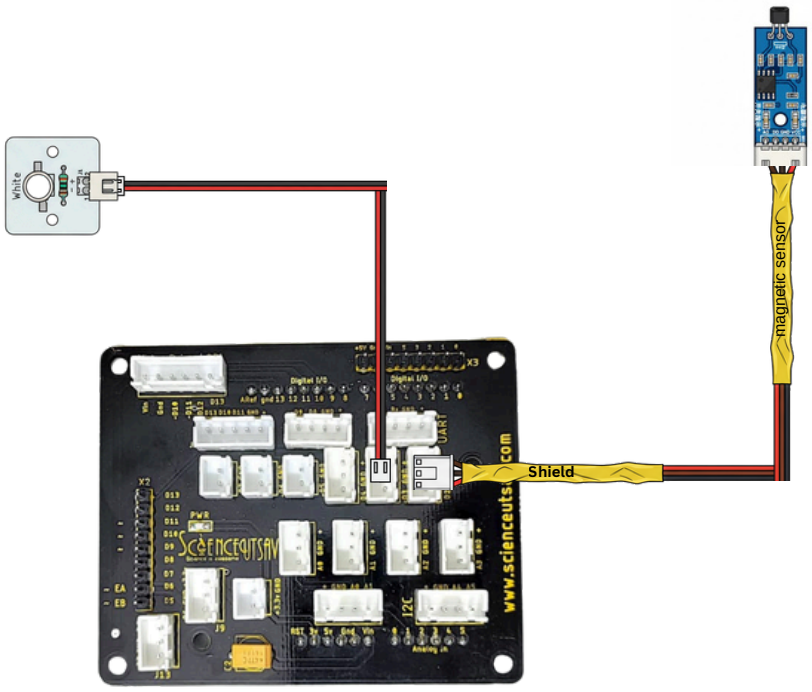
How to Connect

Use RMC wires to connect :

- 1 × HallEffect (Magnet) Sensor→ Arduino pin D2 and GND
- 1 × white 1 Watt SMD LED→ Arduino pin D3 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to gethe code file for this project.”



Code

```
#define HALL_PIN 2  // Hall sensor output pin
#define LED_PIN 3  // LED pin

void setup() {
 pinMode(HALL_PIN, INPUT);
 pinMode(LED_PIN, OUTPUT);
}

void loop() {
 int hallState = digitalRead(HALL_PIN);

 if (hallState == LOW) {    // Magnet detected (output LOW)
  digitalWrite(LED_PIN, HIGH); // Turn ON LED
 } else {
  digitalWrite(LED_PIN, LOW); // Turn OFF LED
 }
}
```

Code Explanation

- In this Arduino program, a Hall sensor is connected to pin 2 (HALL_PIN) and an LED is connected to pin 3 (LED_PIN).
- The setup() function runs once when the Arduino starts. It sets the Hall sensor pin as INPUT and the LED pin as OUTPUT using pinMode().
- The loop() function runs continuously. It reads the Hall sensor value using digitalRead() and stores it in hallState.
- If a magnet is detected (LOW signal), the LED turns ON. If no magnet is present (HIGH), the LED turns OFF.

So overall, the LED glows when a magnet is detected by the Hall sensor and turns OFF otherwise.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Magnetic Detect Basics but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define HALL_PIN 2     // Hall sensor output pin
#define LED1_PIN 3     // First LED on D3
#define LED2_PIN 4     // Second LED on D4

void setup() {
  pinMode(HALL_PIN, INPUT);
  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);
}

void loop() {
  int hallState = digitalRead(HALL_PIN);

  if (hallState == LOW) {          // Magnet detected
    digitalWrite(LED1_PIN, HIGH); // LED1 ON
    digitalWrite(LED2_PIN, HIGH); // LED2 ON
  } else {
    digitalWrite(LED1_PIN, LOW);  // LED1 OFF
    digitalWrite(LED2_PIN, LOW);  // LED2 OFF
  }
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn

the basic idea of Magnetic Detect Basics how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Intermediate: Magnetic Light Indicator

Class Explanation

In this class, students learn A Magnetic Light Indicator system uses a magnetic sensor to detect the presence of a magnet and control an LED. When a magnet comes close, the sensor activates the LED, providing a visual indication. It is commonly used in door sensors and position detection systems.

Materials Required

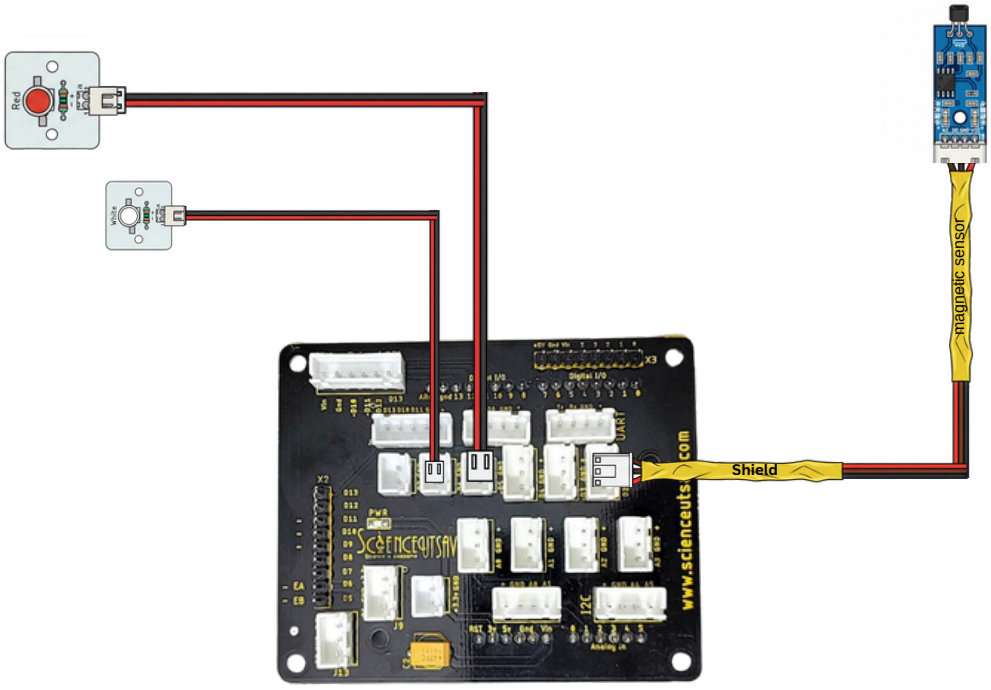
- Arduino with attached shield
- 1 × Hall effect sensor
- 1 × white 1 Watt SMD LED on breakout board
- 1 × Red 1 Watt SMD LED on breakout board
- RMC wires
- USB cable

How to Connect

- Use RMC wires to connect :
- 1 × Hall effect sensor→ Arduino pin D2 and GND
- 1 × white 1 Watt SMD LED → Arduino pin D6 and GND
- 1 × RED 1 Watt SMD LED→ Arduino pin D5 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
#define HALL_PIN 2 // Hall sensor output pin
#define LED_PIN1 6 // LED1 pin
#define LED_PIN2 5 // LED2 pin (opposite behavior)

void setup() {
 pinMode(HALL_PIN, INPUT);
 pinMode(LED_PIN1, OUTPUT);
 pinMode(LED_PIN2, OUTPUT);
}

void loop() {
 int hallState = digitalRead(HALL_PIN);
 if (hallState == LOW) {     // Magnet detected (output LOW)
  digitalWrite(LED_PIN1, HIGH); // Turn ON LED at D6
  digitalWrite(LED_PIN2, LOW); // Turn OFF LED at D5 (opposite)
 } else {
  digitalWrite(LED_PIN1, LOW); // Turn OFF LED at D6
  digitalWrite(LED_PIN2, HIGH); // Turn ON LED at D5 (opposite)
 }
}
```

Code Explanation

- In this Arduino program, a Hall sensor is connected to pin 2 (HALL_PIN), and two LEDs are connected to pins 6 (LED_PIN1) and 5 (LED_PIN2). The second LED works in the opposite way to the first LED.

- The setup() function runs once when the Arduino starts. It sets the Hall sensor pin as INPUT and both LED pins as OUTPUT using pinMode().

- The loop() function runs continuously. It reads the Hall sensor value using digitalRead() and stores it in hallState.

- If a magnet is detected (LOW signal), the LED on pin 6 turns ON and the LED on pin 5 turns OFF. If no magnet is present (HIGH), the LED on pin 6 turns OFF and the LED on pin 5 turns ON.

So overall, the Arduino detects a magnet using the Hall sensor and controls two LEDs in opposite ways.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Magnetic Light Indicator with Arduino but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define HALL_PIN 2   // Hall sensor output pin

#define LED1_PIN 6   // LED1 (same group)
#define LED2_PIN 5   // LED2 (opposite group)
#define LED3_PIN 4   // LED3 (same as LED1)
#define LED4_PIN 3   // LED4 (same as LED2)

void setup() {
  pinMode(HALL_PIN, INPUT);

  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);
  pinMode(LED3_PIN, OUTPUT);
  pinMode(LED4_PIN, OUTPUT);
}

void loop() {
  int hallState = digitalRead(HALL_PIN);

  if (hallState == LOW) {  // Magnet detected
    // Group 1 LEDs ON
    digitalWrite(LED1_PIN, HIGH);
    digitalWrite(LED3_PIN, HIGH);

    // Group 2 LEDs OFF
    digitalWrite(LED2_PIN, LOW);
    digitalWrite(LED4_PIN, LOW);

  } else {
    // Group 1 LEDs OFF
    digitalWrite(LED1_PIN, LOW);
    digitalWrite(LED3_PIN, LOW);

    // Group 2 LEDs ON
    digitalWrite(LED2_PIN, HIGH);
    digitalWrite(LED4_PIN, HIGH);
  }
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learnctive:

the basic idea of Magnetic Light Indicator with Arduino how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Advanced: Magnetic Alert System

Class Explanation

In this class, students learn the A Magnetic Alert System uses a magnetic sensor to detect the presence or absence of a magnetic field and trigger an alert. When a magnet is moved or detected, the system activates an LED or alarm, making it useful for security, door monitoring, and safety applications.

Materials Required

- Arduino with attached shield
- 1 × Tilt sensor
- 1 × Buzzer
- 1 × White 1 Watt SMD LED on breakout board
- RMC wires
- USB cable

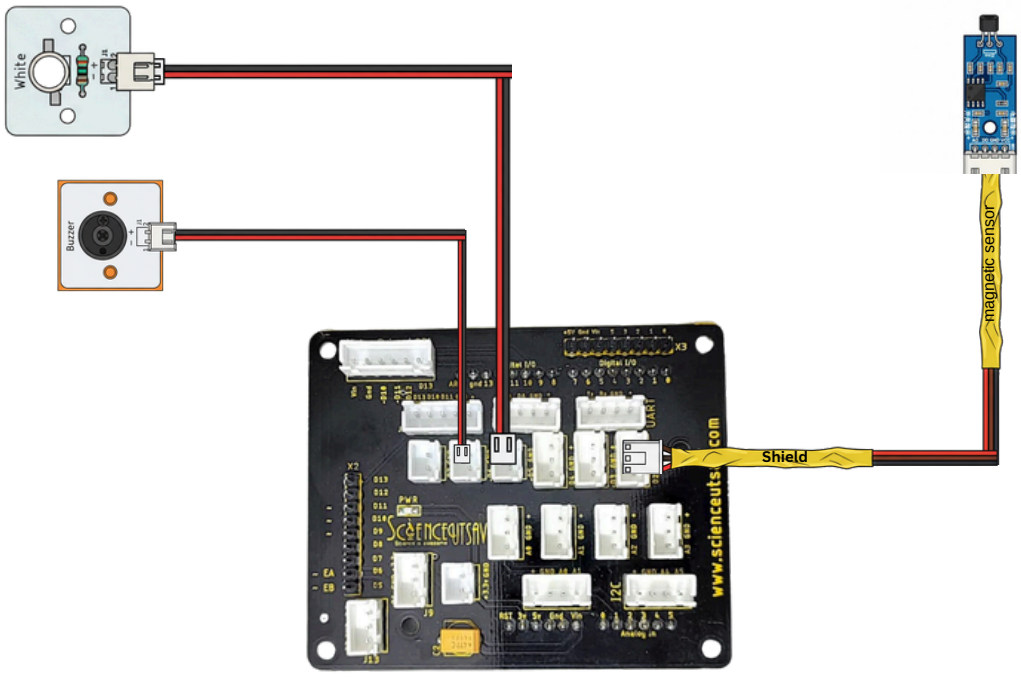
How to Connect

Use RMC wires to connect :

- 1 × Tilt sensor sensor→ Arduino pin D2 and GND
- 1 × Buzzer→ Arduino pin D6 and GND
- 1 × White 1 Watt SMD LED → Arduino pin D5 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
// Hall Sensor with Buzzer and LED (opposite behavior)
// Board: Arduino UNO (or compatible)

#define HALL_PIN 2  // Hall sensor output pin
#define BUZZ_PIN 6  // Buzzer pin (active on detection)
#define LED_PIN 5  // LED pin (opposite behavior)

void setup() {
 pinMode(HALL_PIN, INPUT);
 pinMode(BUZZ_PIN, OUTPUT);
 pinMode(LED_PIN, OUTPUT);

 digitalWrite(BUZZ_PIN, LOW);
 digitalWrite(LED_PIN, HIGH); // Default: LED ON when no magnet
}

void loop() {
 int hallState = digitalRead(HALL_PIN);

 if (hallState == LOW) {     // Magnet detected (output LOW)
  digitalWrite(BUZZ_PIN, HIGH); // Turn ON buzzer
  digitalWrite(LED_PIN, LOW);  // Turn OFF LED (opposite)
 } else {
  digitalWrite(BUZZ_PIN, LOW); // Turn OFF buzzer
  digitalWrite(LED_PIN, HIGH); // Turn ON LED (opposite)
 }
}
```

Code Explanation

- In this Arduino program, a Hall sensor is connected to pin 2 (HALL_PIN), a buzzer to pin 6 (BUZZ_PIN), and an LED to pin 5 (LED_PIN). The LED works in the opposite way to the buzzer.
- The setup() function runs once when the Arduino starts. It sets the Hall sensor as INPUT and both the buzzer and LED as OUTPUT using pinMode(). Initially, the buzzer is turned OFF and the LED is turned ON.
- The loop() function runs continuously. It reads the Hall sensor value using digitalRead() and stores it in hallState.
- If a magnet is detected (LOW signal), the buzzer turns ON and the LED turns OFF. If no magnet is present (HIGH), the buzzer turns OFF and the LED turns ON.
- So overall, the Arduino detects a magnet and activates the buzzer while turning OFF the LED, and does the opposite when no magnet is present.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Magnetic Alert System but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define HALL_PIN 2    // Hall sensor output pin
#define BUZZ1_PIN 6   // Buzzer 1
#define BUZZ2_PIN 7   // Buzzer 2

void setup() {
  pinMode(HALL_PIN, INPUT);

  pinMode(BUZZ1_PIN, OUTPUT);
  pinMode(BUZZ2_PIN, OUTPUT);

  digitalWrite(BUZZ1_PIN, LOW);
  digitalWrite(BUZZ2_PIN, LOW);
}

void loop() {
  int hallState = digitalRead(HALL_PIN);

  if (hallState == LOW) {  // Magnet detected
    digitalWrite(BUZZ1_PIN, HIGH); // Buzzer 1 ON
    digitalWrite(BUZZ2_PIN, HIGH); // Buzzer 2 ON
  } else {
    digitalWrite(BUZZ1_PIN, LOW);  // Buzzer 1 OFF
    digitalWrite(BUZZ2_PIN, LOW);  // Buzzer 2 OFF
  }
}
```

Obective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learnctive:

the basic idea of Magnetic Alert System how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Expert: Hall Sensor Alert System

Class Explanation

A Hall Sensor Alert System uses a Hall effect sensor to detect magnetic fields and generate an alert signal. When a magnet is brought near or removed, the sensor responds by activating an LED or alarm, making it useful for security systems, position sensing, and automated detection applications.

Materials Required

- Arduino with attached shield
- 1 × Tilt sensor
- 1 × Servo motor
- RMC wires
- USB cable

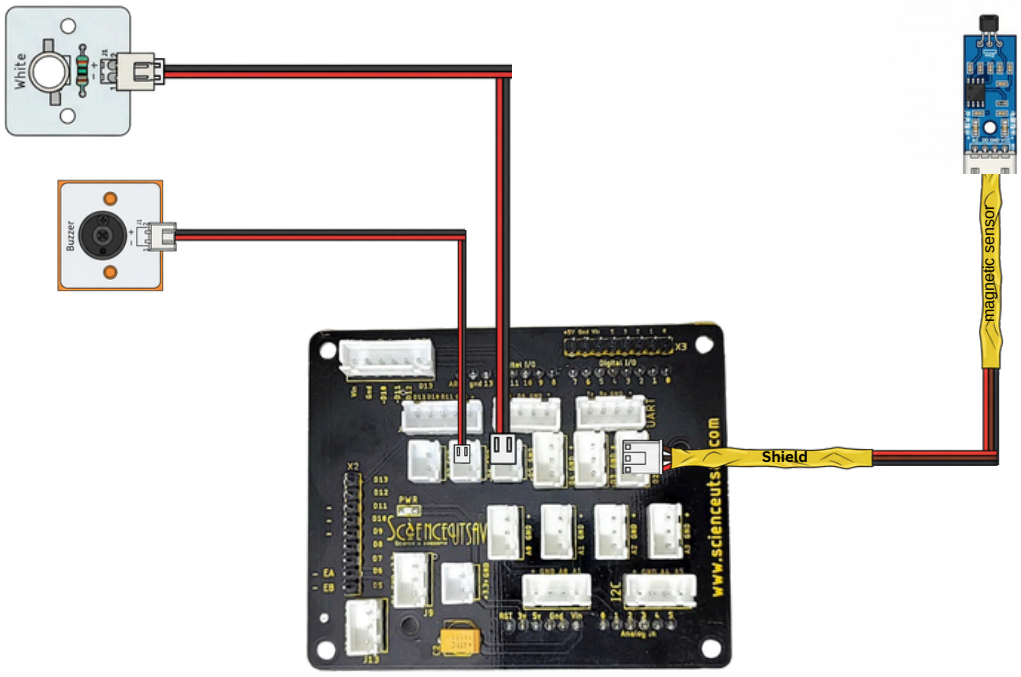
How to Connect

Use RMC wires to connect :

- 1 × Tilt sensor → Arduino pin D2 and GND
- 1 × Servo motor → Arduino pin D3 and GND

Then connect the Arduino to your computer with the USB cable.

“Refer to the link or QR at the back of the book to get the code file for this project.”



Code

```
#define HALL_PIN 2   // Hall sensor output pin

#define BUZZ1_PIN 6  // Buzzer 1
#define BUZZ2_PIN 5  // Buzzer 2

void setup() {
  pinMode(HALL_PIN, INPUT);

  pinMode(BUZZ1_PIN, OUTPUT);
  pinMode(BUZZ2_PIN, OUTPUT);

  digitalWrite(BUZZ1_PIN, LOW);
  digitalWrite(BUZZ2_PIN, LOW);
}

void loop() {
  int hallState = digitalRead(HALL_PIN);

  if (hallState == LOW) {  // Magnet detected
    digitalWrite(BUZZ1_PIN, HIGH); // Buzzer1 ON
    digitalWrite(BUZZ2_PIN, HIGH); // Buzzer2 ON
  } else {
    digitalWrite(BUZZ1_PIN, LOW);  // Buzzer1 OFF
    digitalWrite(BUZZ2_PIN, LOW);  // Buzzer2 OFF
  }
}
```

Code Explanation

- In this Arduino program, a Hall sensor is connected to pin 2 (HALL_PIN), and two buzzers are connected to pins 6 and 5 (BUZZ1_PIN and BUZZ2_PIN).

- The setup() function runs once when the Arduino starts. It sets the Hall sensor as INPUT and both buzzers as OUTPUT using pinMode(). Initially, both buzzers are turned OFF.

- The loop() function runs continuously. It reads the Hall sensor value using digitalRead() and stores it in hallState.

- If a magnet is detected (LOW signal), both buzzers turn ON. If no magnet is present (HIGH), both buzzers turn OFF.

So overall, the system activates both buzzers when a magnet is detected and turns them OFF otherwise.

Try This Yourself Code

This Try This Yourself code keeps the same topic of Hall Sensor Alert System but changes the pattern, values, or outputs slightly so students can practice the same idea with more confidence.

Try This Yourself Code

```
#define HALL_PIN 2    // Hall sensor output pin

#define BUZZ1_PIN 6   // Buzzer 1
#define BUZZ2_PIN 7   // Buzzer 2

#define LED1_PIN 5    // LED 1
#define LED2_PIN 4    // LED 2

void setup() {
  pinMode(HALL_PIN, INPUT);

  pinMode(BUZZ1_PIN, OUTPUT);
  pinMode(BUZZ2_PIN, OUTPUT);

  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);

  // Default state (no magnet)
  digitalWrite(BUZZ1_PIN, LOW);
  digitalWrite(BUZZ2_PIN, LOW);

  digitalWrite(LED1_PIN, HIGH);
  digitalWrite(LED2_PIN, HIGH);
}

void loop() {
  int hallState = digitalRead(HALL_PIN);

  if (hallState == LOW) {  // Magnet detected
    // Buzzers ON
    digitalWrite(BUZZ1_PIN, HIGH);
    digitalWrite(BUZZ2_PIN, HIGH);
```

```
// LEDs OFF (opposite behavior)
 digitalWrite(LED1_PIN, LOW);
 digitalWrite(LED2_PIN, LOW);

 } else {
 // Buzzers OFF
 digitalWrite(BUZZ1_PIN, LOW);
 digitalWrite(BUZZ2_PIN, LOW);

 // LEDs ON
 digitalWrite(LED1_PIN, HIGH);
 digitalWrite(LED2_PIN, HIGH);
 }
}
```

Objective:

- students practice the same topic with a small variation instead of starting from zero
- students compare the first code and the Try This Yourself code to understand what changed and why
- students build confidence by modifying a familiar circuit and logic flow

What Students Learn:
the basic idea of Hall Sensor Alert System how the code connects with the circuit behavior how to upload and test a simple Arduino sketch

Learning Outcomes
After completing this session, students should be able to:

- Understand how magnetic sensors detect the presence of a magnetic field and trigger outputs like LEDs or alarms.
- Learn how magnetic detection is used to control an LED, providing visual indication when a magnet is present.
- Understand how magnetic sensors activate alerts when a magnet is detected or removed, useful for safety and security applications.
- Learn how a Hall effect sensor detects magnetic fields and generates alert signals to control electronic outputs efficiently.

Try Out Other Things

- use the Try This Yourself code after each main class code before asking students to build a completely new variation
- ask students to point out the exact lines that changed between the first code and the Try This Yourself code
- let students first run the Try This Yourself code code as it is and then change one value, one condition, or one output